

Assignment 5: An Unconsidered Vital Signs Report

Oxygen saturation (SpO₂) is a vital sign that measures the percentage of haemoglobin in the blood that is carrying oxygen, providing a quick and non-invasive way to assess how effectively oxygen is being transported to the body's tissues. It is measured using a pulse oximeter, usually placed on a fingertip, and is widely used in clinical and emergency settings to rapidly identify patients with potential respiratory compromise. In healthy adults, normal SpO₂ levels are typically between 95% and 100%, indicating adequate oxygenation. Values between 90% and 94% may suggest mild hypoxaemia and can be associated with conditions such as asthma, pneumonia, or chronic obstructive pulmonary disease (COPD), requiring closer monitoring. A reading below 90% is considered clinically significant hypoxaemia, indicating insufficient oxygen delivery to tissues and the need for urgent clinical intervention, while levels below approximately 85% may indicate severe respiratory failure and can become life-threatening if not treated promptly. In triage settings, SpO₂ is an essential observation used to quickly identify patients at risk of deterioration and support prioritisation of urgent care, particularly when interpreted alongside other vital signs such as respiratory rate and blood pressure to provide a more complete picture of a patient's physiological status. However, clinical interpretation must also consider limitations of pulse oximetry, including reduced accuracy in cases of poor peripheral perfusion, patient movement, or other physiological factors (Hafner et al., 2021).

Reference

Hafner, L., et al. (2021). Pulse oximetry: Understanding its limitations in clinical practice. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9608019/>